

Task 8: ESP32 TRC (Modular Code)

rtc.h

```
main  rtc.cpp  rtc.h §
1 #ifndef RTC_MANAGER_H
2 #define RTC_MANAGER_H
3
4 #include <Arduino.h>
5 #include "RTClib.h"
6
7 class RTCManager {
8     public:
9         RTCManager();
10
11         bool begin();
12         String getFormattedTime();
13         String getDate();
14         String getTime();
15         String getDayOfWeek();
16
17     private:
18         RTC_DS1307 rtc;
19         const char* daysOfTheWeek[7] = {
20             "Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"
21         };
22 };
23
24 #endif
```

rtc.cpp

```
main  rtc.cpp  rtc.h

1 #include "rtc.h"
2
3 RTCManager::RTCManager() {}
4
5 bool RTCManager::begin() {
6     if (!rtc.begin()) {
7         Serial.println("Couldn't find RTC");
8         return false;
9     }
10
11     if (!rtc.isrunning()) {
12         Serial.println("RTC is NOT running, setting time to compile time!");
13         rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));
14     }
15     return true;
16 }
17
18 String RTCManager::getFormattedTime() {
19     DateTime now = rtc.now();
20
21     String yearStr    = String(now.year(), DEC);
22     String monthStr   = (now.month() < 10 ? "0" : "") + String(now.month(), DEC);
23     String dayStr     = (now.day() < 10 ? "0" : "") + String(now.day(), DEC);
24     String hourStr    = (now.hour() < 10 ? "0" : "") + String(now.hour(), DEC);
25     String minuteStr  = (now.minute() < 10 ? "0" : "") + String(now.minute(), DEC);
26     String secondStr  = (now.second() < 10 ? "0" : "") + String(now.second(), DEC);
27     String dayOfWeek  = daysOfTheWeek[now.dayOfTheWeek()];
28
29     return dayOfWeek + String(", ") + yearStr + "-" + monthStr + "-" + dayStr +
30         " " + hourStr + ":" + minuteStr + ":" + secondStr;
31 }
32
33 String RTCManager::getDate() {
34     DateTime now = rtc.now();
35     String yearStr    = String(now.year(), DEC);
36     String monthStr   = (now.month() < 10 ? "0" : "") + String(now.month(), DEC);
37     String dayStr     = (now.day() < 10 ? "0" : "") + String(now.day(), DEC);
38
39     return yearStr + "-" + monthStr + "-" + dayStr;
40 }
```

```

41
42 String RTCManager::getTime() {
43     DateTime now = rtc.now();
44     String hourStr = (now.hour() < 10 ? "0" : "") + String(now.hour(), DEC);
45     String minuteStr = (now.minute() < 10 ? "0" : "") + String(now.minute(), DEC);
46     String secondStr = (now.second() < 10 ? "0" : "") + String(now.second(), DEC);
47
48     return hourStr + ":" + minuteStr + ":" + secondStr;
49 }
50
51 String RTCManager::getDayOfWeek() {
52     DateTime now = rtc.now();
53     return String(daysOfTheWeek[now.dayOfTheWeek()]);
54 }

```

main.ino

```

1 #include "rtc.h"
2
3 RTCManager rtcManager;
4
5 void setup() {
6     Serial.begin(115200);
7     if (!rtcManager.begin()) {
8         Serial.println("RTC init failed!");
9         while (1) delay(10);
10    }
11 }
12
13 void loop() {
14     Serial.println(rtcManager.getFormattedTime());
15     delay(1000);
16 }
17

```